IEEE INFOCOM 2027 | Workshop

Fusion of Data, Operation, Information, and Communication Technologies for Industrial Internet and Society 5.0

The 2nd DOICT-IndSoc Workshop | Call for Papers

May 24, 2027 | Honolulu, Hawaii, USA



The Second International Workshop on Fusion of Data, Operation, Information, and Communication Technologies for Industrial Internet and Society 5.0 (DOICT-IndSoc 2027) will be held in conjunction with IEEE INFOCOM 2027. The workshop focuses on AI and communications, blockchain, digital twins, autonomous driving, industrial control, human-machine collaboration, and related topics.

IEEE INFOCOM is a CCF A / CORE A* conference in computer networking and communications. Building on the first workshop, which attracted submissions from 20 countries and regions, DOICT-IndSoc 2027 will continue to promote interdisciplinary exchange and academia-industry collaboration, exploring new technologies, architectures, and applications for Industrial Internet and Society 5.0.

Topics of Interest

- Architectures for DOICT fusion in Industrial Internet
- Communication networks for industrial automation and cyber-physical systems
- Service-based architectures and sustainable technologies for Industrial Internet
- Cross-domain and multi-resource management for industrial applications
- Edge intelligence and information security for Industrial Internet
- Network support for Human-AI Agents-device collaboration and cooperative autonomy
- Blockchain-based trust, supply chain, and decentralized identity management
- Related autonomous driving technologies in Intelligent Transportation System
- Connected robots/devices/vehicles, and industrial IoT platforms
- Payment channel networks and incentive design for multi-party collaboration
- Software and data compliance and assurance in Industrial Internet
- Time-series forecasting for operation and maintenance in Industrial Internet
- AI model verification and interpretability for industrial intelligence
- AI model privacy and data confidentiality protection in industrial environments
- Digital twin-driven resource management and task scheduling in smart manufacturing
- Human-AI Agents-device collaboration and cooperative autonomy
- AR/VR applications and related technologies in Industrial Internet and Society 5.0

SUBMISSION GUIDELINES

Papers must use the standard IEEE two-column format and must not exceed **six pages, including references**. All submissions will undergo peer review. Accepted papers presented by an author at the workshop will be published in the IEEE INFOCOM 2027 proceedings and IEEE Xplore.

IMPORTANT DATES

Submission deadline	**Dec. 29, 2026**
Acceptance notification	**Feb. 2, 2027**
Camera-ready deadline	**Feb. 16, 2027**
Workshop date	**May 24, 2027**

Organizing Committee

Steering Committee

Hiroshi Esaki, The University of Tokyo, Japan
Manabu Tsukada, The University of Tokyo, Japan
Kensuke Fukuda, National Institute of Informatics, Japan
Megumi Kaneko, National Institute of Informatics, Japan
Celimuge Wu, University of Electro-Communications, Japan
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Workshop website: https://indsocws-pixel.github.io/doict-indsoc/

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